

Configuration backup , os change , backup restore and file transfer

```
ssh -o KexAlgorithms=+diffie-hellman-group14-sha1 -o HostKeyAlgorithms=+ssh-rsa  
admin@192.168.10.1
```

pc to switch

```
scp -T -O -o KexAlgorithms=+diffie-hellman-group14-sha1 -o HostKeyAlgorithms=+ssh-rsa  
.\c1100-universalk9_ias.16.09.04.SPA.bin admin@192.168.10.1:flash:
```

switch to pc

```
scp -T -O -o KexAlgorithms=+diffie-hellman-group14-sha1 -o HostKeyAlgorithms=+ssh-rsa  
admin@192.168.10.1:flash:c1100-universalk9_ias.16.09.04.SPA.bin .\
```

! 1. Enable SSH

ip domain name local.network

crypto key generate rsa modulus 2048

ip ssh version 2

! 2. Enable the Secure Transfer Servers

ip scp server enable

ip ssh server sftp <-- (Only works on modern IOS XE releases)

! 3. Configure AAA and User Privileges (Mandatory for SCP/SFTP Managing Software Files
(Firmware / .bin)

```
Switch(config)# aaa new-model
```

```
Switch(config)# aaa authentication login default local
```

```
Switch(config)# aaa authorization exec default local
```

Pushing an OS image to the router (from PC):

scp C:\path\to\image.bin [admin@192.168.10.1:bootflash:](#)

Verifying and applying the OS image (on Router):

Router# verify /md5 bootflash:image.bin

Router# configure terminal

Router(config)# no boot system

Router(config)# boot system bootflash:image.bin

Router(config)# exit

Router# write memory

Router# reload

Powershell command

Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0

Start-Service sshd

Set-Service -Name sshd -StartupType 'Automatic'

R1(config)# ip domain name local.network

R1(config)# crypto key generate rsa

R1(config)# ip ssh version 2

R1(config)# ip scp server enable

R1(config)# line vty 0 4

R1(config-line)# login local

R1(config-line)# transport input ssh

R1(config-line)# exit

R1(config)# username admin privilege 15 secret Nepal@2026

R1# copy running-config scp://admin@10.0.0.50/backups/Router01.cfg

Backing up a configuration (Router pushing to PC):

Router# copy running-config scp://admin@10.0.0.50/Desktop/backup.cfg

Restoring a configuration (Router pulling from PC):

Router# copy scp://admin@10.0.0.50/Desktop/backup.cfg running-config

username admin privilege 15 secret YourPassword
R1(config)# username admin privilege 15
secret Nepal@2026

R1# copy running-config scp://admin@10.0.0.50/backups/Router01.cfg

scp -O -T -o KexAlgorithms=+diffie-hellman-group14-sha1 -o HostKeyAlgorithms=+ssh-rsa -o
ciphers=+aes128-cbc admin@192.168.10.1:bootflash:c1100-universalk9_ias.16.09.04.SPA.bin
C:\Users\Admin\Desktop\wi

Push file to router (os) with scp

R1# show flash: (for memory kati chha verify garni)

Open Command Prompt, PowerShell, or your Linux/mac terminal on your PC. Use the following syntax to push the file to the router's flash memory:

scp C:\Downloads\c2900-universalk9-mz.SPA.157-3.M2.bin admin@192.168.10.1:flash:

Verify file

```
R1# verify /md5 flash:c2900-universalk9-mz.SPA.157-3.M2.bin
```

TFTP method

To copy ios from pc

```
R1# copy tftp: flash:
```

Address or name of remote host []? 192.168.10.100 <-- (Your PC's IP Address)

Source filename []? c2900-universalk9-mz.SPA.157-3.M2.bin

Destination filename [c2900-universalk9-mz.SPA.157-3.M2.bin]? (Press Enter to keep the same name)

Copy running config from the router to pc

```
R1# copy running-config tftp:
```

Address or name of remote host []? 192.168.10.100 <-- (Your PC's IP Address)

Destination filename [r1-config]? r1-backup.cfg

USB Flash Drive

- Format a USB drive to FAT16 or FAT32 on your PC and copy the IOS .bin file onto it.
- Plug the USB drive into the router's front or rear USB port. You will see a console message indicating a USB device was inserted.
- Copy the file directly to flash:

```
R1# copy usbflash0:c2900-universalk9-mz.SPA.157-3.M2.bin flash:
```

Copy running config from the router to usb

```
show file systems
```

```
R1#copy flash:c1100-universalk9_ias.16.09.04.SPA.bin usb0:
```

```
R1#copy usb0:c1100-universalk9_ias.16.09.04.SPA.bin flash:
```

FTP (For Very Large IOS-XE Files)

Configure your PC's FTP credentials on the router:

```
R1(config)# ip ftp username pc-user
```

```
R1(config)# ip ftp password pc-password
```

```
R1(config)# exit
```

```
R1# copy ftp://192.168.10.100/c2900-universalk9-mz.SPA.157-3.M2.bin flash:
```

Copy from router to pc

```
R1# copy running-config ftp:
```

```
Address or name of remote host []? 192.168.10.100
```

```
Destination filename [r1-config]? r1-backup.cfg
```

```
Writing r1-backup.cfg !
```

```
2103 bytes copied in 1.104 secs (1904 bytes/sec)
```

In case accidentally mero router udyo kasari backup file upload garnin

USB Drive

```
Router# copy usbflash0:r1-backup.cfg startup-config
```

Reload the device so it boots up using the restored configuration:

```
Router# reload
```

TFTP

```
Router# copy tftp: startup-config
```

```
Address or name of remote host []? 192.168.1.100
```

```
Source filename []? r1-backup.cfg
```

```
Destination filename [startup-config]? (Press Enter)
```

```
Router# reload
```

Configure the Archive Features

```
R1(config)# archive
```

```
R1(config-archive)# path scp://admin:password@192.168.10.50/backups/$h-$t.cfg
```

```
R1(config-archive)# exit
```

```
$h hostname
```

```
$t timestamp
```

Create the Kron Policy List

```
R1(config)# kron policy-list BACKUP-POLICY
```

```
R1(config-kron-policy)# cli archive config
```

```
R1(config-kron-policy)# exit
```

Run on the 1st of every month at 02:00 AM

```
R1(config)# kron occurrence BACKUP-1ST at 2:00 1 recurring
```

```
R1(config-kron-occurrence)# policy-list BACKUP-POLICY
```

```
R1(config-kron-occurrence)# exit
```

Run on the 15th of every month at 02:00 AM

```
R1(config)# kron occurrence BACKUP-15TH at 2:00 15 recurring
```

```
R1(config-kron-occurrence)# policy-list BACKUP-POLICY
```

```
R1(config-kron-occurrence)# exit
```

Verification

show kron schedule: Displays the upcoming scheduled times for your backup policies.

show archive: Shows the defined archive path, variables, and a list of the most recent successful backups

debug kron all: Useful for troubleshooting if your scheduled backups fail to trigger.

The "Safe Mode" (Revert Timer)

If you are configuring a router remotely and accidentally change an IP address or firewall rule that locks you out, you lose access to fix it. The archive feature provides a dead-man's switch.

Router# configure terminal revert timer 10

If your changes work perfectly, you permanently apply them and stop the timer by typing:

Router# configure confirm

Running vs. Startup

Router# show archive config differences

If you formatted the flash: memory and accidentally deleted the actual IOS operating system (the .bin file) rather than just the configuration, the device will be stuck in rommon > mode. Let me know if this is the case, as restoring the IOS requires a different recovery procedure.

Preparation

Connections: You need two cables connected at the same time:

- A Console cable (to type commands).

- An Ethernet cable connecting your PC directly to the router's primary network port (usually GigabitEthernet0/0).

PC Setup: Give your PC a static IP address (e.g., 192.168.1.100 with subnet mask 255.255.255.0).

TFTP Server: Open your TFTP application on your PC, and ensure the correct Cisco IOS .bin file is in its root directory.

rommon 1 > IP_ADDRESS=192.168.1.1

rommon 2 > IP_SUBNET_MASK=255.255.255.0

rommon 3 > DEFAULT_GATEWAY=192.168.1.100

rommon 4 > TFTP_SERVER=192.168.1.100

rommon 5 > TFTP_FILE=c2900-universalk9-mz.SPA.157-3.M2.bin

Initiate the TFTP Download

rommon 6 > tftpdnld

Type y. You will see a string of !!!!! as the file transfers. This can take several minutes depending on the file size.

Boot the Router

rommon 7 > boot